Department of Electrical, Electronic and Computer Engineering
BET Hons: Electrical Engineering

Development of a Web-Based Community Energy Monitoring Dashboard with Comparative Analytics Using Simulated Smart Circuit Controller Data

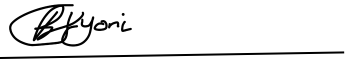
THESIS

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DECLARATION

I, Bongani Nyoni, declare that the contents of this thesis represent my own unaided work, and that the thesis has not previously been submitted for academic examination towards any qualification. Furthermore, it represents my own opinions and not necessarily those of the Cape Peninsula University of Technology.



Signed

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Abstract

Household electricity readings are increasingly available from smart meters and smart circuit controllers, yet raw values of voltage, current and power seldom show whether consumption is rising, which load dominates, when demand peaks or how a home compares with similar homes. This thesis presents the design, implementation and technical evaluation of a web-based community energy monitoring dashboard that turns circuit-level readings into comparative analytics. Because access to a commercial smart circuit controller application programming interface (API) was not available, a controlled dataset was simulated for ten anonymised sites, six controller channels and 30 days at 15-minute resolution (172 800 readings), modelling a generic multi-channel smart circuit controller with known data-quality faults and abnormal-use events injected under experimenter control. A layered pipeline validates, stores and aggregates the readings in a PostgreSQL relational database, and eight transparent algorithms compute energy, estimated cost, peak demand, load ranking, baseline and peer ratios, data completeness and abnormal-use warnings.

All 240 injected invalid records were correctly rejected by the validation layer. Independent recomputation, written separately from the production pipeline, matched its output to within floating-point precision. Short-gap interpolation reduced the mean monthly energy error against ground truth from 1.63% to 1.33%. On a separately generated hold-out dataset, a two-stage warning rule raised precision from 0.49 to 1.00 at the cost of recall, which fell from 0.63 to 0.40, showing that the rule trades sensitivity for reliability rather than improving both. The pipeline was further run, unmodified except for a new controller-channel type, against ten real households from the public Low Carbon London smart-meter dataset, confirming that the pipeline generalizes beyond the simulator it was built around and revealing that whole-community comparison becomes substantially more misleading as real household variance grows. Within-group peer comparison, specified in the design but not yet implemented in the live dashboard, is identified as the most important remaining piece of future work, alongside the participant-based usability study defined for this project but not yet completed.

Keywords: CBi-like monitoring capability, smart energy monitoring, community energy dashboard, comparative analytics, smart metering, IoT, demand-side management, energy feedback, POPIA, web-based dashboard.

Mathematical Notations

V - RMS voltage measured at the controller channel, in volts.

I - RMS current measured at the controller channel, in amperes.

PF - Power factor of the monitored load.

P - Active power consumed by a monitored load, in kilowatts.

E - Electrical energy consumed over a selected period, in kilowatt-hours.

Δt - Sampling interval used for energy integration (0.25 h for the 15-minute simulated grid).

C - Estimated cost of consumption for a selected tariff, in rand.

T - Configurable illustrative electricity tariff, in rand per kilowatt-hour.

η - Data completeness: the share of expected readings that are valid.

Definitions

Glossary of items

Algorithm - A step-by-step procedure for solving a problem using a computer

API (Application Programming Interface) - A set of rules and protocols for building and interacting with software applications, allowing different systems to communicate.

CBi Astute - A specific brand and model of smart energy monitoring device capable of measuring electrical parameters at the household level.

Dashboard - A visual interface displaying key performance indicators and data insights, often using charts, graphs, and gauges.

Database Schema - The structural definition of a database, including tables, fields, relationships, and constraints.

Community energy monitoring - The combined monitoring of several households, rooms, buildings or selected loads so that collective consumption patterns can be understood.

Comparative Analytics - The process of comparing data sets across different entities (e.g., households) to identify patterns, outliers, and benchmarks.

Demand-side management - Actions that help users reduce, shift or better understand electricity demand instead of only increasing supply.

Nomenclature

AI - Artificial Intelligence

AMI - Advanced Metering Infrastructure

API - Application Programming Interface

CC-BY - Creative Commons Attribution (open licence)

CSS - Cascading Style Sheets

CSV - Comma-Separated Values

DEL - Domestic Electrical Load

FN - False Negative

FP - False Positive

IoT - Internet of Things

IQR - Interquartile Range

ISO - International Organization for Standardization

KPI - Key Performance Indicator

kWh - Kilowatt-hour

NILM - Non-Intrusive Load Monitoring

PDF - Portable Document Format

PF - Power Factor

POPIA - Protection of Personal Information Act

RMS - Root Mean Square

SUS - System Usability Scale

TP - True Positive

Wi-Fi - wireless local area networking

XAI - Explainable Artificial Intelligence

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Chapter 1: Introduction

1.1 Introduction

Electricity use in a household is often treated as a monthly bill rather than a measurable operating pattern. That creates a gap between the amount of energy data that can be collected and the amount of insight that ordinary users can act on. In South Africa, this gap matters because households rely on electricity for cooking, refrigeration, heating, lighting, communication devices, security systems and small business activity, while electricity cost and peak demand remain practical constraints. Access to mains electricity in South Africa rose from 76.7% of households in 2002 to 90.2% in 2024 (Statistics South Africa, 2025), confirming that electricity dependence is now widespread enough for household-level monitoring to have practical value.

Smart meters and smart circuit controllers make household electricity use more visible, but visibility on its own is not enough. A reading of power or energy does not by itself say whether the household is improving, whether a load is dominating consumption, whether a repeated evening peak is forming, or whether a site is above the average profile of a comparable group. Smart circuit controllers with CBi-like capability provided the practical reference point for this project because they combine energy monitoring, scheduling, switching control, load management and automation using variables such as current, voltage, time and power (CBI-electric, 2026). The useful engineering layer sits between those readings and user interpretation: data validation, storage, aggregation, comparison, visualisation and privacy control. This thesis documents the design, construction and technical evaluation of that layer.

This project developed a web-based community energy monitoring dashboard using simulated smart circuit controller data. The shift to simulation removed dependency on a specific commercial device or API while keeping the data close to the operating logic of CBi Astute-type controllers. The simulated dataset mimics realistic timestamped readings such as voltage, current, active power, energy consumption, device category, switching state and data-quality flags, and the same architecture was subsequently validated, unmodified, against a real public smart-meter dataset (Chapter 4), which demonstrates that the simulation-first approach did not lock the system to synthetic data alone.

1.2 Background and Problem Statement

1.2.1 Engineering significance of the interpretation gap

An energy-monitoring system must distinguish between measurement, information and decision support. Measurement produces electrical quantities at particular instants; information emerges when those quantities are assigned to a circuit, time interval and site; decision support requires a meaningful reference against which a user can interpret the resulting pattern. A dashboard that displays a power reading without context may be technically correct while remaining operationally unhelpful. Conversely, an attractive visualisation built from incomplete or invalid measurements may communicate an incorrect conclusion with considerable confidence. The engineering problem is

therefore not merely a matter of chart selection. It is a chain-of-evidence problem in which every displayed conclusion depends on the integrity of the preceding transformations.

The distinction between power and energy is particularly important. Power describes the instantaneous rate of electrical energy transfer, whereas energy accumulates power over time. A relatively high-power cooking appliance may operate briefly and contribute less monthly energy than a lower-power appliance operating continuously. A resident who confuses the highest observed power with the largest energy contributor could prioritise the wrong load. Circuit-level breakdowns, period totals and peak-demand indicators should consequently be interpreted together rather than as interchangeable measurements. The proposed system makes this distinction explicit by retaining channel identity, integrating readings across a known sampling interval and reporting peak demand separately from total consumption.

A community dimension introduces another analytical difficulty: sites are heterogeneous. Household size, appliance ownership, occupancy patterns and electric water-heating arrangements can affect consumption independently of energy efficiency. A raw ranking may therefore compare unlike circumstances. This thesis uses an anonymised leave-one-out community mean as an implemented initial benchmark while acknowledging that the comparison is descriptive rather than a fair efficiency score. Profile-matched comparisons are treated as a design recommendation, not as a feature already delivered. This separation between demonstrated capability and future improvement is essential for a defensible evaluation.

Household electricity readings are often difficult to interpret when they are viewed only as isolated values, raw exports or spreadsheet-style records. A user may see voltage, current, active power or accumulated energy, yet these readings do not automatically show whether consumption is improving, which monitored load contributes most to usage, when peak demand occurs, or how one household compares with an anonymised group profile.

The research problem addressed by this thesis is the lack of a privacy-aware web dashboard that can import structured smart-controller energy readings, validate and store them, apply comparative analytics and present meaningful individual and community-level energy insights. Since direct access to a specific commercial smart-device API was not available, the prototype used a controlled simulated smart circuit controller dataset that mimics realistic circuit-level energy measurements, and was later cross-checked against an independent, publicly available real-meter dataset. The core problem was the absence of a clear interpretation layer that turns energy readings into baseline, peak-load, device-ranking, estimated-cost and comparative information.

South Africa adds a specific operating context. Smart metering is moving into the formal electricity system, with Eskom's smart meter programme having started in 2025 and expected to continue until 2029 (Eskom, 2026). Even where formal utility metering improves, a household or small community still needs local interfaces that explain consumption patterns, highlight peaks and compare behaviour in practical terms. The dashboard therefore sits below utility-grade infrastructure and focuses on awareness, analytics and interpretation rather than revenue-grade measurement.

1.3 Research Aim and Objectives

1.3.1 Aim

The aim of the research was to design, develop and evaluate a web-based community energy monitoring dashboard that uses simulated smart circuit controller data to provide comparative analytics for household or small-community electricity consumption, covering data ingestion, validation, storage, baseline comparison, peak detection, device ranking, estimated cost and community comparison.

1.3.2 Research Objectives

This study was guided by the following five objectives:

1. Data specification: define the electrical variables and dataset structure required to support a smart circuit controller-based community energy monitoring dashboard, including a formal data specification covering voltage, current, active power, cumulative energy, timestamp, controller channel identifier, device category and switching state, with value ranges, units and data-quality thresholds.
2. Data pipeline: design and implement a secure data pipeline that ingests, validates, stores and pre-processes smart circuit controller energy readings for dashboard consumption, with schema-enforced database tables, explicit validation rules, missing-value handling and an audit log.
3. Analytics development: develop comparative analytics algorithms to derive interpretable energy metrics, including consumption baselines, community comparisons, peak-demand detection, device-level rankings, anomaly warnings and estimated cost calculations, each verified against a hand-calculated reference case.
4. Dashboard prototype: build a responsive, web-based dashboard prototype that presents individual and community-level energy information through interactive visualisations and exportable summaries, rendering correctly across desktop and mobile viewport widths.
5. Evaluation: evaluate the dashboard prototype against criteria of calculation accuracy, data completeness, responsiveness and abnormal-use warning performance, and define the usability evaluation instrument for the remaining participant-based study.

1.3.3 Research Questions

1. Which electrical variables and metadata fields are required to represent smart circuit controller energy data in a dashboard-ready format?
2. How can simulated smart circuit controller readings be generated so that they are realistic enough for testing ingestion, validation, analytics and visualisation?
3. How can raw timestamped readings be structured into a reliable data model for household and community analytics?
4. Which comparative metrics are most useful for identifying high usage, peak periods, abnormal consumption and monitored-load ranking?
5. How can a web dashboard present energy feedback clearly enough for users to interpret consumption patterns without reading raw data tables?
6. What privacy and security controls are required for an architecture that may later connect to real smart-controller or household energy data?

1.4 Hypothesis

H1: A web-based dashboard that uses structured simulated smart circuit controller data and comparative analytics improves the interpretability of household and community energy consumption patterns compared with viewing isolated raw readings or spreadsheet-style data.

H0: A web-based dashboard using structured simulated smart circuit controller data and comparative analytics does not materially improve the interpretability of household and community energy consumption patterns compared with viewing isolated raw readings or spreadsheet-style data.

This thesis establishes the technical precondition for testing H1 and H0, a working, accuracy-verified pipeline, a dashboard, and reports the quantitative evaluation results achieved to date. The participant-based interpretation study needed to accept or reject the hypothesis directly is identified in Chapter 5 as the principal remaining item of future work.

1.5 Motivation

1.5.1 Research contribution and practical relevance

The research contributes an integrated and inspectable engineering workflow rather than a new electrical sensor. Its principal contribution lies in showing how readings can be validated, reconstructed where defensible, aggregated and compared without concealing uncertainty introduced by missing data. The study also demonstrates why technical evaluation must examine both numerical correctness and the meaning of a displayed metric. A ratio can be computed exactly and still lead to an unhelpful inference if the comparison population is poorly matched. This distinction informs the discussion of external validity and peer-group design in the concluding chapter.

At the practical level, a browser-based interface can reduce the burden of inspecting long spreadsheets or isolated device screens. A resident may need to know whether a geyser or cooking circuit dominates a particular day, whether a repeated evening peak is emerging and whether recent consumption differs from the household's own prior pattern. Such questions can be answered from a modest set of interpretable measures if the underlying time-series pipeline is reliable. The prototype is not claimed to deliver measured energy savings; that outcome would require an intervention study with participants and a suitable comparison condition. Instead, the completed research establishes and evaluates the technical infrastructure that would make such a study possible.

The motivation is both technical and practical. Technically, the project builds a complete engineering chain from smart-controller-style energy data representation to dashboard interpretation, drawing on electrical quantities, time-series data, databases, web systems, analytics and privacy engineering. Practically, the project responds to the need for South African households and small communities to understand consumption behaviour more clearly, especially where energy cost and peak usage affect daily decisions.

A simulated smart-controller dataset was not a downgrade once the research was framed correctly: it gave the prototype controlled input conditions, making it possible to test whether the dashboard handles expected readings, abnormal readings, missing intervals, switching-state changes and comparison cases against a known ground truth. The project additionally validated the same pipeline against a real public smart-meter dataset (Chapter 4), which was not required by the original proposal but substantially strengthens the claim that the architecture generalises beyond the simulator it was designed around.

1.6 Assumptions

- A controlled simulated smart circuit controller dataset was used for the primary prototype testing and development.
- The simulated dataset contains realistic variables: timestamp, voltage, current, active power, interval energy, cumulative energy, device category, switching state and a data-quality flag.
- The prototype did not require direct access to a commercial smart-device API.

- The architecture was kept data-source agnostic, which was subsequently confirmed by running it, unmodified except for a new channel type, against a real public dataset.
- The dashboard was not used for certified billing, tariff enforcement, utility switching or direct grid control.
- Estimated cost calculations use a configurable illustrative tariff and are not presented as an official bill.
- Community comparison uses pseudonymised identifiers and does not expose household names, addresses or detailed occupancy routines.
- Forecasting and machine-learning-based explanation remained optional extensions and were not required for the core comparative dashboard.

1.7 Thesis Structure

Chapter 1 establishes the research context, problem, aim, objectives, hypothesis and assumptions. Chapter 2 reviews South African energy conditions, smart metering, smart circuit controller capability, energy feedback and comparative analytics, dashboard design and data processing, dataset simulation, privacy and related deployed systems. Chapter 3 presents the system design and implementation that was actually built, covering the architecture, data specification, data model, algorithms, dashboard interface, implementation tools and testing approach. Chapter 4 presents the technical evaluation results: data-quality handling, calculation accuracy, comparative analytics outputs, abnormal-use warning performance, pipeline performance and responsiveness, and validation against a real public smart-meter dataset. Chapter 5 discusses the results, states the limitations encountered, and concludes with recommendations for future work.

Chapter 2: Literature Review

2.1 South African Residential Energy Context

2.1.1 Contextual implications for dashboard requirements

South African residential electricity use is shaped by the interaction of household demand, supply constraints, tariff arrangements and differing access to efficient appliances. These factors motivate a design that can represent energy in physical units independently of the price model applied to it. Energy totals expressed in kilowatt-hours remain comparable across tariff configurations, whereas rand-denominated estimates depend on a selected tariff and the applicable billing rules. Accordingly, the dashboard presents an illustrative configurable cost rather than treating a single flat-rate multiplication as a certified municipal or utility bill.

The relationship between energy monitoring and demand-side management also requires care. Monitoring can reveal the timing and magnitude of consumption, but it does not itself shift load or reduce peak demand. Behavioural changes, equipment replacement or automatic control would be needed to realise an intervention. The present system remains observational and analytic. Its hourly heatmap and peak-demand indicators identify candidate periods for further investigation; they do not demonstrate that any demand-response action occurred. This is consistent with the scope established in Chapter 1 and avoids conflating the potential usefulness of feedback with experimentally measured savings.

The South African household energy context supports the need for practical monitoring tools. Increased access to electricity has expanded the number of households using electrical appliances, and greater dependence makes consumption behaviour more important to understand. The 2024 General Household Survey shows that 90.2% of households were connected to mains electricity in 2024, compared with 76.7% in 2002 (Statistics South Africa, 2025). The consequence is not only improved access; it is also a stronger need for visibility at household and community level.

Smart metering is becoming part of the formal electricity system. Eskom states that its smart meter programme started in 2025 and will continue until 2029 (Eskom, 2026), confirming that smarter measurement is part of the current national direction. A utility rollout, however, does not automatically solve local interpretation: users still need interfaces that explain consumption patterns, highlight peak periods and compare behaviour in a way that is meaningful at household scale.

South African load-profile research also supports the value of comparison. Toussaint and Moodley (Toussaint and Moodley, 2020) describe the Domestic Electrical Load Metering Hourly dataset as a major South African household electricity database and use clustering to derive representative daily load profiles, showing that consumption behaviour can be grouped into archetypes. Ritchie, Engelbrecht and Booysen (Ritchie, Engelbrecht and Booysen, 2022) further demonstrate the value of data-driven household electricity profiles by synthesising statistically representative demand profiles from South African smart meter data. Neither study solves the dashboard problem directly, but both

support the logic that household energy data can be structured, compared and interpreted when the correct analytical layer is used — the logic this thesis implements and tests.

2.2 Smart Metering, CBi Astute Data and IoT Data Acquisition

2.2.1 Measurement provenance and device independence

A reliable ingestion architecture should describe not only what a numerical field means but also where it originated. A reported active-power value may be measured directly by a controller, derived from voltage and current, or calculated from interval energy. These alternatives can have different error characteristics. Where voltage, current and power factor are supplied, the relation between them provides a useful consistency test, subject to measurement tolerance and time alignment. Where an external smart-meter dataset supplies energy alone, imposing a voltage-current consistency check would be inappropriate. Source-specific validation capabilities must therefore be recorded explicitly rather than assuming that every source contains every electrical field.

Device independence is achieved by translating each source into a canonical record with stable site and channel identifiers, timestamps, units and quality indicators. The adapter is responsible for mapping source fields and native intervals; the downstream analytics should operate on the normalised representation. This separation is useful when moving from a simulated controller export to a half-hourly public smart-meter dataset. The reported external-data run supports the feasibility of this design, although it does not establish compatibility with every vendor API, authentication method or proprietary payload format. A future hardware integration would still require tests of clock drift, packet loss, device restart behaviour and any limitations imposed by the vendor.

Smart metering and consumer IoT devices both collect electrical measurements but serve different roles. Utility smart meters are normally part of formal advanced metering infrastructure (AMI) and billing systems, while consumer smart devices support local monitoring, control and automation. The dashboard in this research sits on the consumer-device side: it was not intended to certify billing, but to use accessible device data to support energy awareness and comparison.

The CBi Astute Smart Controller is relevant because it combines monitoring, scheduling, load management and automation. CBi states that the device is Wi-Fi enabled, supports scheduled energy management, provides energy monitoring and can respond to variables such as current, voltage, time and power (CBI-electric, 2026). These variables are sufficient for a dashboard prototype because power and energy can support trend analysis, cost estimation, peak detection and baseline comparison, which is why the simulated dataset built for this project (Chapter 3) was designed to mirror exactly this variable set rather than a hardware-specific template.

A technically sound design cannot assume unrestricted access to a private vendor cloud. Because a specific commercial API was not available for this project, the dataset was simulated rather than imported, with the architecture kept able to accept CSV import or an approved API in future. This kept the project feasible without depending on unofficial reverse engineering, and the resulting source-

agnostic design is what later made it possible to ingest a real, independent public dataset through the same pipeline without code changes to the validation or analytics layers (Chapter 4).

2.3 Energy Feedback and Comparative Analytics

2.3.1 Feedback design, behavioural interpretation and fairness

Energy-feedback research indicates that presentation choices influence whether consumption information is useful. Karlin, Zinger and Ford (Karlin, Zinger and Ford, 2015) synthesised 42 feedback studies and reported an overall positive but heterogeneous relationship between feedback and conservation, with frequency, medium, comparison message and intervention duration among the relevant moderators. Their findings support a dashboard that can provide repeated, interpretable feedback; they do not imply that this specific prototype has already changed household behaviour. A future user study would need to test interpretation accuracy separately from any longer-term consumption outcome.

Social comparison is potentially informative because it gives an otherwise abstract energy total a reference point. However, a mean is affected by the spread of the underlying distribution, and a small number of high-consuming households can pull the average upward. For a site i, a leave-one-out mean excludes that site's own energy from the reference population, reducing the mechanical dependence between the numerator and benchmark. It does not correct differences in dwelling characteristics or appliance ownership. This limitation is important when interpreting the wider ratio range observed in the real London sample. A neutral status band and an own-history baseline provide additional context, but cannot replace careful selection of genuinely comparable peers.

An appropriate future comparison system could first group sites using predeclared characteristics or load-profile features, then compute a leave-one-out mean or robust median within each group. Group sizes should remain large enough to protect privacy and to avoid unstable statistics. The decision to create groups must also be assessed for bias: a cluster derived from energy use may reflect occupancy and equipment as much as efficiency. For this reason, the thesis recommends describing peer results as relative consumption patterns rather than as definitive judgments about household performance.

Energy feedback research is the strongest scholarly foundation for this topic. Darby (Darby, 2006) argues that feedback makes energy visible and supports learning about how consumption can be controlled. Fischer (Fischer, 2008) adds that effective feedback depends on frequency, duration, content, breakdown, medium, presentation, comparisons and combination with other instruments. This directly supports a dashboard approach, because the dashboard is not only a display; it is the mechanism that organises frequency, comparison, breakdown and presentation.

Reviews of household intervention studies find that feedback is one of the more effective tools for reducing consumption, particularly when it is frequent and specific (Abrahamse et al., 2005), (Ehrhardt-Martinez, Donnelly and Laitner, 2010). Ehrhardt-Martinez, Donnelly and Laitner (Ehrhardt-Martinez, Donnelly and Laitner, 2010) reviewed advanced metering and feedback programmes and

found that feedback can create household electricity-saving opportunities when it is designed and delivered effectively. Karlin, Zinger and Ford (Karlin, Zinger and Ford, 2015) later showed through meta-analysis that the effect of energy feedback depends heavily on design and context, implying that the dashboard had to be evaluated as a feedback system and not merely as a database viewer.

Comparative analytics gives the dashboard its distinctive value. Allcott (Allcott, 2011) found that social comparison reports reduced average household energy consumption by about 2% in a large field experiment. Schultz et al. (Schultz et al., 2007) showed that descriptive social norms alone can produce a boomerang effect, in which low-consuming households increase their usage toward the norm, and that pairing the norm with an approving or disapproving cue prevents this. These two findings shaped the design choices actually carried through into the implementation described in Chapter 3: every comparison shown to a resident is paired with the household's own baseline, and the result is reported as a neutral status band rather than a numeric leaderboard position, specifically to avoid a boomerang effect among already-efficient households.

2.4 Web-Based Dashboard Design and Data Processing

2.4.1 Data quality as part of visual communication

A chart inherits the quality limitations of its source data. For example, a line chart with an unmarked communication gap can appear to show a reduction in consumption when the system merely failed to receive measurements. Likewise, filling a long outage with interpolated values may create a visually smooth curve at the expense of factual credibility. A robust interface should therefore make completeness and imputation status visible alongside the energy estimate. This thesis distinguishes short, bounded interpolation from long gaps deliberately retained as missing, and it evaluates the effect of that policy against simulator ground truth.

Aggregation creates another source of potential misinterpretation. An hourly mean of power, an hourly peak and an hourly total of energy answer different questions. A stacked channel breakdown should use the same time window for all categories, while a peer comparison should use the same duration and compatible units for the focal site and the reference population. When the interval changes from fifteen minutes to thirty minutes, energy integration must use the new duration rather than assuming a fixed multiplier. These considerations justify the canonical time-series model and independent recomputation strategy used in the implemented system.

A web dashboard is suitable because it can be accessed from different devices, supports centralised data storage and can visualise trends using familiar charts. The interface should prioritise interpretation over technical clutter: users should see the current consumption position, the main contributors to energy use, the peak period, the cost estimate and the comparison result without needing to inspect raw data.

The data processing layer is as important as the interface. Raw readings may contain missing timestamps, repeated values, communication gaps, out-of-range values or device-state changes. The

system must therefore validate timestamps, remove duplicates, flag missing intervals, aggregate readings into usable periods and preserve audit trails. Non-Intrusive Load Monitoring (NILM) literature is relevant but not central to this prototype: Hart (Hart, 1992) established NILM as a method of identifying appliance loads from aggregate measurements, but because simulated smart-controller channels (and the real dataset validated in Chapter 4) are labelled directly at source, this project did not need appliance-level disaggregation.

The dashboard architecture separates data source, ingestion, validation, storage, analytics and visualisation, which is what made it possible, in practice, to move from simulated CSV data to a real public dataset (Chapter 4) without rewriting the analytics or presentation layers.

2.5 Smart-Controller Dataset Simulation and Controlled Validation

2.5.1 Relationship to public household electricity datasets

The UK-DALE dataset documented by Kelly and Knottenbelt (Kelly and Knottenbelt, 2015) combines whole-house demand with appliance-level measurements from five United Kingdom homes. Its value for this thesis is methodological: it demonstrates how timestamps, meter identities and appliance labels can support analyses beyond a single aggregate total. The REFIT dataset documented by Murray, Stankovic and Stankovic (Murray, Stankovic and Stankovic, 2017) provides a further example, covering twenty households with whole-house and appliance-level readings collected at high temporal resolution. REFIT also describes the importance of cleaning and documenting missing observations. These published datasets establish precedents for structured, labelled residential load records, but their measurement conditions and household populations differ from the simulated South African-oriented controller scenario.

The simulation approach offers one advantage that observational datasets do not always provide: the experimenter can retain a complete, known reference series before faults are injected. It then becomes possible to distinguish an algorithm's arithmetic error from the energy error caused by missing readings. However, simulator-generated schedules cannot capture the full variety of occupant decisions, appliance faults, seasonal effects and electrical measurement noise. The separate Low Carbon London validation is therefore a meaningful extension because it challenges the same pipeline with external consumption patterns. Neither the simulator nor a small external sample should be treated as representative of all South African households.

The simulated smart-controller dataset is central to this project. It is not random numbers without engineering meaning: it was generated from site profiles, controller channels, monitored-load categories, time-of-day schedules, weekday/weekend patterns and controlled abnormal events. This allowed the dashboard to be tested against realistic operating logic while keeping the data source independent of a commercial API.

Household load datasets from the literature informed the design of this simulator even though none were used directly as the primary dataset. Public datasets such as UK-DALE (Kelly and Knottenbelt, 2015) and REFIT (Murray, Stankovic and Stankovic, 2017) show how whole-house and appliance-level readings can be structured as labelled time series, supporting the decision to label each simulated controller channel directly by load category. Controlled validation is stronger than pretending the data is live: known input conditions allowed calculation of exact expected totals, peaks, ratios and flags, against which the implemented pipeline could be checked (Chapter 4).

2.6 Privacy, Security and Ethical Handling of Smart Energy Data

2.6.1 Threat model and data minimisation

Privacy requirements should be expressed as specific risks and controls. Fine-grained electricity data can reveal recurring periods of activity, unusual absences and appliance-use signatures. Even where names are replaced by codes, repeated time-series patterns can be distinctive, so pseudonymisation is not equivalent to complete anonymisation. The prototype's separation of resident and administrator views reduces casual disclosure, while the aggregation of community comparisons limits the amount of individual information exposed. These are architectural safeguards rather than proof that every possible inference attack has been eliminated.

A production deployment would require authenticated access, role-based authorisation at the API and database layers, secure transport, credential management, retention limits and an auditable process for responding to access requests. The rule that a comparison is displayed only when the group contains at least three sites is a minimum suppression policy, not a formal guarantee that released aggregates satisfy k-anonymity in all circumstances. Repeated queries or differencing between overlapping groups could still disclose information. The legal and ethical review must therefore examine both direct identifiers and what can be inferred from repeated access to apparently aggregated data.

Smart energy data is not neutral. It can reveal when people are at home, when appliances are used and whether a household follows predictable routines. Asghar et al. (Asghar et al., 2017) show that smart meter data can expose private information, while Beckel et al. (Beckel et al., 2014) show that household characteristics can be inferred from energy data. Even though the prototype used simulated data, the system was designed as if real household data could be connected in future.

South African law strengthens this requirement. The Protection of Personal Information Act (POPIA) sets out conditions for lawful processing, including accountability, processing limitation, purpose specification, further processing limitation, information quality, openness, security safeguards and data subject participation (Republic of South Africa, 2013). The dashboard design therefore needed privacy controls at the architecture level, not only in an ethics form: site identifiers are pseudonymised throughout (shown only as, for example, Site A to Site J, or anonymised household codes), community views show aggregated comparisons rather than household names, and raw data access is separated from dashboard views. The k-anonymity model (Sweeney, 2002), in which a released record cannot

be distinguished from at least k − 1 others, motivated the rule implemented in Chapter 3 that a comparison is shown only when the comparison group contains at least k = 3 sites.

2.7 Anomaly Detection in Smart Meter Data

2.7.1 Interpretable detection and threshold selection

Statistical warning rules are attractive for a prototype because their decision logic can be inspected. The interquartile range summarises the middle half of an observed distribution, and a Tukey-style upper fence provides a conventional way to identify unusually large values. For energy time series, however, an unusual value is not necessarily an error or waste: a holiday gathering, colder weather or an extended cooking period may produce legitimate high consumption. Detection should therefore be framed as a prompt for investigation rather than as a diagnosis of faulty equipment or misconduct.

A second-stage severity requirement can reduce false alerts, but it inevitably excludes some events near the threshold. This is precisely the trade-off shown by the hold-out results in Chapter 4. Precision measures how many raised alerts correspond to injected events; recall measures how many injected events were found. Reporting both prevents a zero-false-positive outcome from being mistaken for universal detection. Future evaluation should also consider event duration, alert burden per household-month and threshold stability across seasonal periods, rather than relying on a single accuracy-like score.

Anomaly-detection research for electricity consumption spans a spectrum from heavyweight AI methods to lightweight statistical rules. Himeur et al. (Himeur et al., 2021) survey AI-based anomaly detection for building energy use and find that most published methods rely on supervised or deep-learning models trained on richly labelled, sub-metered data, which real household deployments seldom have. Al-Ghaili et al. (Al-Ghaili et al., 2021) review anomaly detection specifically for advanced metering infrastructure and note that fine time granularity makes anomalies easier for a utility to catch, but that most of the techniques they cover target data manipulation and theft rather than ordinary abnormal-use events. Closer to the approach used in this project, Jurj et al. (Jurj et al., 2021) apply an interquartile-range outlier rule to building electricity data and find it a practical, interpretable baseline, though sensitive to seasonal variation in the underlying distribution.

Taken together, this literature motivated the lightweight approach used in this project: a per-channel Tukey fence (Tukey, 1977) over daily and night-time totals (Chapter 3, Section 3.4) is far simpler than the deep models most reviews favour, trading some detection power for transparency and a threshold a non-expert user can sanity-check. The quantitative evaluation in Chapter 4 measures that trade-off directly rather than assuming it away.

2.8 Existing Monitoring Platforms

Several deployed systems demonstrate the dashboard half of this problem without the validation half. Bagde et al. (Bagde et al., 2026) built a web-based IoT dashboard that visualises real-time and periodic electrical data with interactive charts to help users spot peak-usage periods, but, like most deployed dashboards, it reports visualisation without documenting data validation, quantitative pipeline accuracy or peer comparison. Muhammed et al. (Muhammed et al., 2022) report a similar IoT home-energy-monitoring prototype with real-time display in both wireless and offline modes, and Mohamed Ariff et al. (Mohamed Ariff et al., 2025) add real-time monitoring to a smart-home energy-management system; neither paper quantifies validation accuracy, reports a gap-handling strategy, or compares a household against its peers.

This is the gap this thesis targets directly: not a new way to visualise consumption, but a documented, testable pipeline between the circuit-level reading and the comparison shown on screen, with every stage of that pipeline quantitatively evaluated in Chapter 4.

2.9 Literature Review Conclusion

The literature supports the need for a dashboard that does more than display readings. South African energy conditions create a practical need for household-level awareness, while smart meter rollouts show that measurement is becoming more important in the national electricity environment. CBi-like smart circuit controller capability gave this project a practical device class to mimic without making the prototype dependent on that device. Feedback scholars show that users need clear, frequent and comparative information, and that comparison must be designed carefully to avoid a boomerang effect. Household energy datasets show that time-series load profiles can be structured and analysed. Deployed dashboard systems show visualisation without quantified validation, while anomaly-detection reviews show a trade-off between detection power and transparency that this project resolves in favour of transparency. Privacy scholars and POPIA show that household energy data must be handled carefully. The research gap addressed by this thesis is therefore a practical engineering gap: a privacy-aware web dashboard that turns structured smart-controller-style readings, validated first in simulation and then against real data, into comparative community energy insight, with every pipeline stage quantitatively evaluated rather than assumed to work.

Chapter 3: Research Methodology and System Design

3.1 Monitoring and Analytics Scheme

3.1.1 Design rationale and separation of concerns

The layered architecture assigns a clear responsibility to each processing stage. The source layer creates or receives readings without deciding how the dashboard should present them. Ingestion maps source-specific fields into a consistent schema. Validation checks whether the record is usable and records the reason when it is not. Gap handling aligns observations to a defined sampling grid and applies the restricted interpolation policy. Persistent storage maintains readings, summaries and audit information. Analytics calculates derived measures, while the presentation layer communicates those measures to the intended audience.

This separation supports reproducibility because a result can be traced backwards through identifiable transformations. For example, a displayed daily total should be linked to the channel readings included in the period, their quality flags, the native interval duration and any imputation applied. The same principle supports debugging: if a chart differs from an independent reference, the investigation can determine whether the discrepancy arose during import, unit conversion, filtering, aggregation or display. Modularity also reduces the risk that replacing a data source requires rewriting validated numerical algorithms.

The monitoring scheme begins with structured simulated smart-controller readings and ends with dashboard feedback. The core variables are timestamp, voltage, current, active power, interval energy, cumulative energy, device category, switching state and a data-quality flag. The system has six layers: data source, ingestion and validation, gap handling, storage, analytics and dashboard. Each layer can be tested on its own, and the data source can change from simulated CSV files to an approved API, a local gateway, or as demonstrated in Chapter 4 an independent public dataset, without changing the later layers.

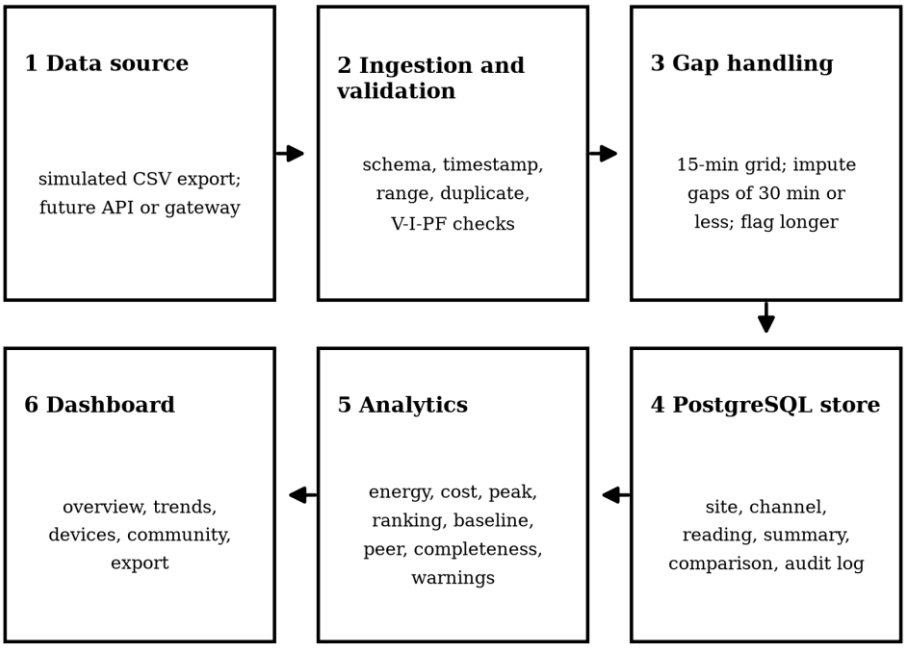


Figure 3.1: Layered architecture of the prototype.

3.2 System Requirements

3.2.1 Requirements traceability and acceptance criteria

Functional requirements should be connected to observable acceptance criteria. Import functionality is meaningful only when malformed and duplicate records are handled deterministically. Daily and monthly graphs are meaningful only when their totals reconcile with stored readings. A cost indicator must disclose its tariff assumption. A community comparison must exclude the focal site from the reference mean and suppress results when the privacy threshold is not satisfied. The existing test strategy maps these concerns to injected-fault checks, independent numerical recomputation, query measurements and responsive-layout checks.

Non-functional requirements are similarly measurable but must be interpreted within the conditions of the test. A browser viewport test at specified widths demonstrates layout behaviour at those widths; it does not guarantee accessibility on every device or assistive technology. Query timings measured on the deployed configuration describe that configuration and dataset size; they do not establish performance at utility-scale volumes. The thesis therefore distinguishes requirements supported by reported tests from qualities that would require further field or load testing.

Functional requirements covered: importing simulated smart circuit controller readings from CSV files; storing readings by timestamp, site, device and measured variable; displaying daily, weekly and monthly consumption graphs; calculating baseline and community comparisons; identifying peak demand periods and high-consumption devices; generating an estimated cost using a configurable tariff; and flagging missing, duplicate and out-of-range readings. Non-functional requirements

covered protecting user identity through pseudonymised identifiers, providing a responsive web interface for desktop and mobile viewing, and keeping the system modular so that the data source could move from simulation to a future smart-controller API, export file, or an independent real dataset without redesigning the analytics and dashboard layers. This last requirement was validated directly: Chapter 4 reports the real-data run that confirms it was met.

3.3 Simulated Smart-Controller Data Specification and Generation

3.3.1 Sampling, units and electrical consistency

The simulator represents each controller channel as a regularly sampled time series. At a fifteen-minute interval, four observations correspond to one hour and ninety-six to a complete twenty-four-hour day. Thirty days across ten sites and six channels therefore yield 30 × 96 × 10 × 6 = 172,800 expected readings before faults are introduced. Stating this count explicitly is useful because it provides a denominator for completeness calculations and a check on the grid-building stage. Missing source rows should not silently reduce the number of expected observations.

The relation P = V × I × PF / 1000 is appropriate for single-phase active power when RMS voltage and current and the applicable power factor refer to the same measurement interval. It also highlights the difference between active power in kilowatts and apparent power in kilovolt-amperes. Multiplying active power by the duration in hours gives interval energy in kilowatt-hours. For a fifteen-minute interval, a constant 2 kW load contributes 0.5 kWh. This simple example serves as a dimensional sanity check for the implementation, but the reported production accuracy is supported by the actual unit tests and independent recomputation described later.

Switching state is a useful contextual variable but should not be assumed to map perfectly to zero consumption. A device can consume standby power, and some circuits aggregate multiple downstream appliances. The simulator's fixed rated-power and power-factor parameters provide controlled physical consistency, yet they simplify actual variation caused by supply voltage, reactive components and transient operating states. Those simplifications should be preserved in the interpretation of the evaluation results rather than generalised to certified metering accuracy.

The dataset was generated using site profiles and controller-channel usage rules. Ten sites were generated (three low-use, four medium-use and three high-use), each with six controller channels — geyser, fridge, lighting, plugs, cooking and background load — over a 30-day period (1 to 30 January 2026) at a 15-minute interval. This gives 2,880 intervals per channel and 172,800 readings in total. Rated power and power factor were fixed per channel category (geyser 3.0 kW / 1.0; fridge 0.15 kW / 0.85; lighting 0.30 kW / 0.98; plugs 0.60 kW / 0.95; cooking 2.4 kW / 1.0; background 0.08 kW / 0.9), and behaviour was shaped by time-of-day schedules, a weekday/weekend split, and random variation. The electrical fields are kept physically consistent through the relations:

$$P = V \cdot I \cdot PF / 1000 \quad (3.1)$$

$$E = \Sigma\, P \cdot \Delta t \quad (3.2)$$

$C = E \cdot T$ *(3.3)*

where V is RMS voltage in volts, I is RMS current in amperes, PF is the power factor, P is active power in kilowatts, Δt = 0.25 h, and T is a configurable illustrative tariff in rand per kilowatt-hour.

Known faults were injected into the simulated export so that the validation layer could be tested against a known ground truth: a 4-hour communication gap at one site (96 readings removed), random loss of 0.3% of readings (550 readings), 200 duplicate re-imported readings, 30 out-of-range voltages, 10 negative power values, and shuffled arrival order. Thirty abnormal-use events were separately injected on randomly chosen site-channel-days: 20 full-day events, where one channel's power for a whole day was scaled by a factor of 1.15 to 3.5, and 10 overnight-only events, where the 00:00-05:00 total was scaled by a factor of 1.2 to 3.0. A second dataset, generated with a different random seed, was reserved purely as a hold-out test set for the warning rules (Chapter 4).

3.4 Data Model and Algorithms

3.4.1 Temporal integrity, missing data and auditability

A time-series database design must distinguish source arrival order from measurement time. Network delivery may be delayed or shuffled, and the same record may be imported more than once. A composite uniqueness rule based on device and measurement timestamp prevents duplicate readings from inflating energy totals. Recording the original quality status and the applicable rule enables a reviewer to explain why an observation was excluded or accepted. The audit log provides a separate record of operational actions, supporting accountability without making it a substitute for data validation.

Interpolation is deliberately limited to short gaps. For a missing point bounded by valid observations, linear interpolation estimates a value between the endpoints; for two consecutive missing points, the same principle divides the interval into equal steps. Such estimates are defensible only under a smooth-change assumption and may be inaccurate when a load switches abruptly. The longer four-hour gap is therefore left unfilled. This policy makes an explicit trade-off between completeness and the risk of manufacturing consumption that was never measured. The evaluation against known ground truth quantifies its effect for the particular simulated conditions, not for every possible load profile.

Data completeness should be calculated against the expected native grid rather than the number of successfully imported rows. The numerator must also be defined consistently: observed-valid completeness and completeness including imputed values are different indicators. Displaying the latter without identifying the former could overstate the quantity of actual measured information. The reported before-and-after completeness figures are best interpreted as complementary indicators of observation coverage and usable reconstructed coverage.

The database uses a relational structure because the data has clear entities: sites, controller channels, readings, daily summaries, comparison results and audit logs. PostgreSQL was used throughout, both for local development and via a managed instance for the deployed system. switching_state and quality_flag are enforced by CHECK constraints at the database level; duplicate and range checks are enforced in the validation layer. Sites are pseudonymised and never shown with a name or address.

Entity	Key fields	Purpose
Site	site_id, data_source, anonymised_label	Groups readings without exposing identity; distinguishes simulated from real sites
Smart controller channel	device_id, site_id, controller_channel, device_category, rated_power, switching_state	Links readings to the monitored controller channel
Reading	reading_id, device_id, timestamp, voltage, current, power, energy_kWh, state, quality_flag	Stores every measurement, flagged or not, under its quality_flag
Daily summary	summary_id, device_id, date, total_kWh, peak_power, cost_estimate	Supports dashboard trend and cost views
Comparison result	comparison_id, site_id, period, baseline_kWh, group_average_kWh, ratio, status_flag	Stores comparative analytics outputs
Audit log	event_id, user_id, action, timestamp	Tracks pipeline and access events for accountability

Six validation rules are applied in evaluation order, the first failure setting the flag on a reading:

R1 – schema presence, enforced before a row reaches the other rules.

R3 – uniqueness of (device, timestamp).

R2 – a valid timestamp on the source's native sampling grid (15 minutes for the simulator; a half-hourly source also satisfies this check when configured to its own interval).

R4 – voltage within 230 V ± 10% (207 V to 253 V), checked only when a source reports voltage.

R5 – $0 \leq P \leq 1.2 \times$ rated power, checked only when a source reports both voltage and current.

R6 – agreement between measured power and V·I·PF within 5%.

Every reading, flagged or not, is kept in the same table under its quality_flag, so nothing is silently discarded. Gaps of up to two native intervals are filled by linear interpolation and flagged as imputed; longer gaps stay missing so the dashboard does not invent consumption.

Eight comparative analytics algorithms were implemented as pure, independently testable functions:

A1 – interval and period energy, by equation 3.2.

A2 – estimated cost, by equation 3.3, using an illustrative flat tariff labelled as an estimate, not a bill.

A3 – peak site demand, the hour with the highest mean demand, and intervals above the 95th percentile.

A4 – channel ranking by total energy.

A5 – baseline comparison against the household's own mean daily energy for weekdays or weekend days over a reference window.

A6 – community comparison: site i against the mean of all other sites for the same period (leave-one-out), shown only when the comparison group contains at least k = 3 sites.

A7 – data completeness η, the share of expected readings that are valid.

A8 – two-stage abnormal-use warnings: stage 1 applies a Tukey (Tukey, 1977) interquartile-range fence to each channel's daily energy totals, with a floor at 15% above the median; stage 2 grades each exceedance by severity and adds a night base-load rule for the 00:00-05:00 window.

A6 is, by design, a whole-community leave-one-out comparison rather than a within-group peer comparison. A within-group (profile-matched) comparison was specified during design as a way to avoid the boomerang effect (Schultz et al., 2007) discussed in Chapter 2, but was not implemented in the live dashboard within the project period; this is recorded honestly in Chapter 4's results and revisited as future work in Chapter 5, rather than being presented as complete.

3.5 Dashboard Interface

3.5.1 Information hierarchy and chart interpretation

The interface prioritises a small number of questions that residents are likely to ask. The current consumption indicator answers how much energy was used over the selected period. The peak-power card identifies the highest demand condition rather than the largest accumulated energy. Device rankings answer which monitored categories contribute most to total use. A time-of-day heatmap helps identify recurring patterns that might be obscured by daily totals. These panels should be read together because no single measure fully describes household consumption.

Chart design also affects comparison validity. A line graph should use a consistent time axis and clearly label kilowatt-hours or kilowatts; a bar chart should not mix peak power with total energy on an unlabeled common scale. A community reference line should indicate whether it represents the whole community or a matched peer group. In the delivered prototype it represents the whole-community comparison. The interface's resident-facing neutral band avoids exposing a named ranking, while the separate administrator view supports operational oversight. This access distinction should remain explicit in any future usability or privacy assessment.

The dashboard is a single responsive page generated from the database, with no third-party scripts. The resident-facing view shows three key performance indicator (KPI) cards (today's consumption, peak power, and status against the whole-community average) and five chart panels: a daily consumption trend line, a device-ranking bar chart of total and peak energy per channel, a line chart comparing the site against the whole-community average, a stacked bar of today's breakdown by channel, and an hourly usage-pattern heatmap by day of week. A separate administrator-only view additionally ranks all households by consumption for operational oversight; residents do not see this ranked view, only their own neutral status band. Data can be exported as CSV or PDF through the browser.

3.6 Implementation Tools

The backend was implemented in Python using the FastAPI framework, with PostgreSQL as the relational database, both locally and via a managed instance for the deployed system. The frontend was implemented in React, built with Vite, and styled with a CSS architecture that supports both a light and a dark presentation via a single theme attribute. Git was used for version control throughout.

3.7 Testing and Evaluation Approach

3.7.1 Verification, validation and generalisability

Verification asks whether the software implements the specified calculation correctly; validation asks whether the model and resulting outputs are appropriate for the intended application. The nineteen unit tests and independent recomputation provide strong evidence for numerical verification within the tested cases. Injected-fault recovery examines whether quality-control rules behave as specified. Hold-out abnormal-event testing probes whether a threshold calibrated on one simulation retains its behaviour on another. The public smart-meter run addresses a different question: whether the pipeline can process a source that was not generated by the simulator.

These tests are complementary rather than interchangeable. Agreement between two scripts does not establish that the input readings represent a real household. A real-data import does not demonstrate that the dashboard improves interpretation. A responsive layout does not prove that users understand the charts. The proposed System Usability Scale and task-based interpretation comparison would address the last point, but their absence means the stated interpretability hypothesis remains untested directly. This separation strengthens the thesis by matching each conclusion to the evidence actually collected.

Evaluation was carried out at three levels: data correctness, system performance and warning quality, with a fourth, user-interpretation level defined but not yet executed. Data-quality handling was checked by comparing flags with the injected faults. Calculation accuracy was checked in two independent ways: unit tests comparing each algorithm with a hand-calculated reference case, and a second, independently written script that recomputed site, device-day, peak and count results directly from the raw CSV without using the production pipeline's own code path, compared against the database to a ±5% target. Abnormal-use warnings were scored at site-channel-day level for precision and recall, first on the calibration dataset and then on a separately generated hold-out dataset. Performance was measured as pipeline run time and the median of 30 executions of each dashboard query. Responsiveness was tested in headless Chromium at 1366, 768, 375 and 320 px, measuring horizontal overflow. The planned usability study uses the System Usability Scale (Brooke, 1996) with at least five participants, who would complete interpretation tasks first on a raw table and then on the dashboard; this study had not been conducted at the time of writing and is carried forward as future work.

3.8 Ethical and Regulatory Considerations

The prototype reduced ethical risk by using simulated data for its primary development and testing: no real household identity, address or behavioural routine was required for the dashboard build. The real-data validation reported in Chapter 4 used an existing, open, CC-BY-licensed public dataset (the Low Carbon London smart-meter trial) rather than newly collected participant data, which avoided the need for new informed-consent procedures while still testing the pipeline against genuinely external data. Should real participant data be connected in future, participation would require informed consent, clear purpose specification, secure storage, role-based access and the right to withdraw, consistent with POPIA (Republic of South Africa, 2013). The dashboard does not perform control actions on electrical loads; its scope is monitoring and analytics only.

3.9 Risks and Mitigation

The principal risk accepted at the design stage was that simulated data might be seen as less realistic than live readings. This was mitigated by using realistic site profiles, injecting known abnormal events, and keeping the architecture source-agnostic; the real-data validation in Chapter 4 is the direct resolution of this risk, since it shows the same pipeline handling genuinely external data without modification to its validation or analytics logic. Other risks managed during the project included the possibility of calculation errors in aggregation, mitigated with independent recomputation and unit tests (Chapter 4); and tariff differences between municipalities, mitigated by presenting cost only as a configurable, clearly labelled estimate.

3.10 Scope and Limitations of the Design

The scope was limited to monitoring, comparative analytics and visualisation using simulated and, subsequently, real public smart-meter data. The work does not replace certified electricity meters, official billing, municipal tariff systems or Eskom smart meter infrastructure. Community comparison in the delivered system is a whole-community average rather than a within-group peer average, which Chapter 4 shows becomes an important source of distortion as household variance grows; this is carried forward explicitly as a limitation rather than minimised.

3.11 Work Plan Outcome

The project followed the work plan set out in the original proposal (literature review and source consolidation; data-source investigation; system requirements and architecture design; database and ingestion-layer development; analytics algorithm development; dashboard interface development; testing with sample data; and final report writing), with the real-data validation against the Low Carbon London dataset (Chapter 4) added once the simulated pipeline was stable, and the usability evaluation remaining as the one planned activity not completed within the project period.

Chapter 4: Results

This chapter reports the technical evaluation results obtained from the implemented system described in Chapter 3. All figures and numbers in this chapter are taken directly from the running system and its evaluation scripts; none are projected or illustrative.

4.1 Reading Fields

Table 4.1 lists the fields carried by every reading processed by the pipeline.

Field	Content and use
site_id	pseudonymised site; no name or address stored
device_id	controller channel identifier
controller_channel	load category: geyser, fridge, lighting, plugs, cooking or background
timestamp	ISO 8601 time with offset; 15-minute grid
voltage_v, current_a	RMS values; range and consistency checks
power_kw	active power; primary input for energy
energy_kwh_interval	energy in the interval
cumulative_energy_kwh	running energy total
switching_state	on, standby or off
quality_flag	valid, imputed, missing, duplicate, out-of-range or abnormal_event

Table 4.1: Reading fields used by the pipeline.

4.2 Data Quality

4.2.1 Interpretation of the fault-handling results

The reported detection of all 240 injected invalid records is a result about the known injected classes: duplicate re-imports, out-of-range voltage and negative power. It should not be interpreted as a universal sensitivity estimate for every possible corrupt measurement. Other fault classes, such as plausible but incorrect sensor readings or synchronized clock drift, could pass the documented checks. The injected missing observations are also different from invalid records because absence is detected by comparison with the expected time grid rather than by examining a received row.

The improvement from 99.63% observed completeness to 99.96% after gap filling illustrates how reconstruction increases usable coverage while retaining the long communication outage as a visible limitation. The eleven isolated losses that were not imputed demonstrate why the term 'isolated' in a

fault-generation procedure does not necessarily guarantee that a missing value has suitable neighbours after all losses are combined. The distinction between observed and reconstructed data should remain visible to users and evaluators.

Every injected invalid record was flagged by the correct rule (Table 4.2). After gridding, 646 expected readings were absent: 550 from random losses and 96 from the communication gap. Of these, 539 isolated gaps were imputed and the 96 readings of the 4-hour gap stayed missing, since gap-filling only bridges runs of up to two missing intervals. Overall completeness was 99.63% before gap-filling and 99.96% after. The lowest site-day completeness value falls on the day of the communication gap, and the dashboard reports this rather than hiding it.

Fault	Injected	Flagged	Rule
Duplicate re-import	200	200	R3
Voltage out of range	30	30	R4
Negative power	10	10	R5
Isolated missing readings	550	539 imputed[1]	grid
4 h communication gap	96	96 missing	grid

[1] 11 of the 550 isolated gaps could not be imputed, either because they sat adjacent to another dropped interval or at the start or end of a device's reading series.

Table 4.2: Injected faults and validation outcome (Dataset A).

4.3 Calculation Accuracy

4.3.1 Sources of numerical and estimation error

The near-zero relative differences between the production pipeline and independently recomputed aggregates support the correctness of the implemented arithmetic under the tested conditions. Floating-point discrepancies of the reported magnitude are not practically significant relative to the ±5% acceptance target. However, numerical agreement should not be confused with recovery of the unobserved physical signal. When readings are missing, two correctly implemented calculations may still differ from the complete ground-truth energy total because information has been lost.

The reported reduction in mean monthly error from 1.63% to 1.33% on the primary dataset is therefore best understood as a benefit of limited gap reconstruction under the simulator's particular load patterns. The remaining error is consistent with leaving the long outage unfilled. A more aggressive method might reduce numerical error for smooth loads but could worsen it for switched loads. The study's conservative interpolation limit is consequently a methodological choice rather than an attempt to maximise apparent completeness at any cost.

All nineteen unit tests passed. The independent recomputation matched the database with a maximum relative error of 4.0×10^{-14} for site-month energy and 3.3×10^{-15} for device-day energy, with zero difference in daily peaks and valid-reading counts across 1,800 device-days — far inside the ±5% target, with the residual attributable to floating-point rounding. Against ground truth, the mean absolute monthly energy error per device was 1.63% (maximum 7.05%) when missing readings were simply excluded, and 1.33% (maximum 6.86%) with short-gap interpolation. The hold-out dataset gave 1.69% and 1.40% (maxima 11.33% and 11.01%). Interpolation therefore removes some of the bias caused by isolated losses, a roughly 18% relative reduction in mean error on both datasets, and most of the remaining error comes from the 4-hour communication gap, which is deliberately not filled because it exceeds the two-interval gap-filling limit.

4.4 Comparative Analytics Outputs

4.4.1 Interpreting load rankings and community ratios

The HOUSE-004 load breakdown demonstrates why circuit-level categories offer a more actionable explanation than a single household total. Cooking and geyser consumption together account for a substantial share of the reported recent-day energy, but the ranking applies to that site's selected day rather than proving that these categories dominate every site or every season. The hourly peak around 18:00 is also consistent with the simulator's evening schedules. It is evidence that the pipeline preserves the intended temporal structure, not an independent discovery about South African household peak behaviour.

The widening of whole-community comparison ratios from the simulated sample to the real London sample is especially important. It indicates that the benchmark's interpretation is sensitive to population heterogeneity even when the ratio is calculated consistently. A value substantially above one identifies consumption above the selected reference mean; it does not by itself identify inefficient behaviour. The results motivate future peer matching and the presentation of group size, spread and baseline context. Such additions would improve interpretability without altering the verified arithmetic of the existing comparison.

The ten simulated households used 12,040.9 kWh over the simulated month, about 1,204.1 kWh per site or 40.1 kWh per site per day. For Site HOUSE-004, the live dashboard's breakdown for its most recent day shows cooking as the largest share (31.4%), followed by the geyser (27.9%), plug loads (21.4%), the fridge (8.7%), lighting (5.7%) and background load (4.9%) (Figure 4.3). Site HOUSE-004 peaked in the 18:00 hour, visible as the darkest column in the heatmap (Figure 4.1), consistent with the evening-use pattern built into the simulator.

The live dashboard compares each site only against the whole-community average, not a within-group peer mean. For the ten simulated households alone (administrator view, Figure 4.2), this ratio ranged from 0.57 to 1.61. Section 4.6 repeats this comparison for the ten real London households, where the much larger variance in real consumption pushes the same ratio far wider (0.39 to 3.14): a whole-community comparison without peer grouping can make a typical high-use household look far more unusual than it is. The leave-one-out within-group peer-group comparison described in Chapter 3 had not yet been implemented in the live dashboard at the time of writing, and this remains future work (Chapter 5).

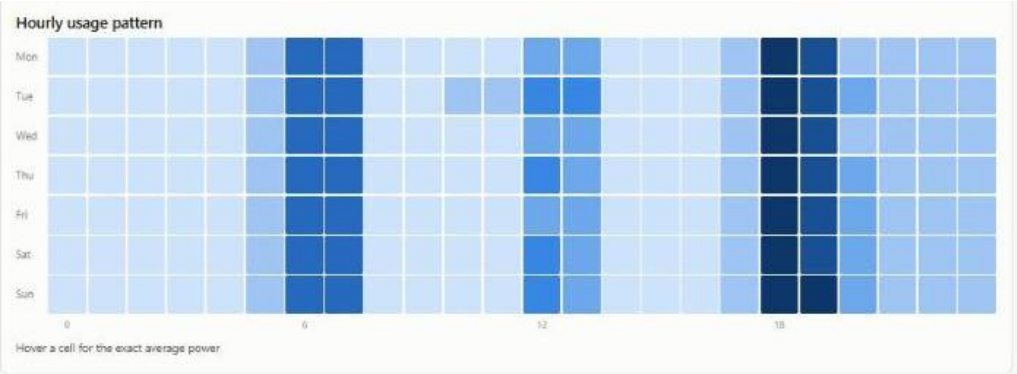


Figure 4.1: Hourly usage-pattern heatmap for Site HOUSE-004 (average power by hour and day of week; darker cells indicate higher average power), from the live dashboard.

HOUSEHOLD	TODAY'S KWH	PEAK KW	VS. COMMUNITY AVERAGE
HOUSE-003	60.03	5.15	161% - Above average
HOUSE-006	59.67	5.21	160% - Above average
HOUSE-009	57.98	5.24	155% - Above average
HOUSE-002	41.50	3.40	106% - Typical
HOUSE-005	40.89	3.31	104% - Typical
HOUSE-008	40.52	3.39	103% - Typical
HOUSE-010	23.98	2.08	58% - Below average
HOUSE-004	23.83	2.10	58% - Below average
HOUSE-007	23.68	1.99	57% - Below average
HOUSE-001	23.60	1.94	57% - Below average

Figure 4.2: Admin ranking view across the ten simulated households, ranked by today's kWh and status versus the whole-community average, from the live dashboard.



Figure 4.3: Load-category breakdown for Site HOUSE-004, from the live dashboard.

4.5 Abnormal-Use Warnings

4.5.1 Precision-recall trade-off and operational consequences

The calibration and hold-out tables show a consistent pattern: requiring a stronger exceedance eliminates the reported false positives while missing more injected events. On the hold-out dataset, stage 1 finds nineteen of thirty events with twenty false positives, whereas stage 2 finds twelve events with no false positives. The resulting precision of 1.00 for stage 2 is conditional on this finite test set; it does not guarantee that a future deployment will never produce a false alarm. Likewise, recall of 0.40 indicates that most injected events were not detected under the stricter rule.

The appropriate threshold depends on the intended use. A resident-facing informational warning may benefit from avoiding frequent false alarms, which can reduce trust and lead users to ignore notifications. A safety-critical monitoring application would instead need a different validation regime and likely a higher-sensitivity detector. The present prototype is not a safety protection device. Its warning is an interpretive aid, and the results should be reported as a measured operating trade-off rather than as evidence of universally superior anomaly detection.

Table 4.3 summarises warning performance across 30 injected day- and night-level events (20 full-day and 10 overnight-only) per dataset, with thresholds calibrated on the calibration dataset only.

In-sample, stage 1 (any exceedance of the IQR fence) found 21 of 30 events but also raised 14 false alerts, mostly from natural day-to-day variation in cooking and plug energy. Severity grading and the night rule removed every false alert, at the cost of missing 5 further borderline events (16 of 30 found). The same pattern held on the hold-out dataset: stage 1 found 19 of 30 events with 20 false alerts, and the two-stage rule found 12 of 30 with zero false alerts. The two-stage rule therefore trades recall for precision rather than improving both: it only raises a warning once an exceedance clears the higher severity threshold or the night floor, which suits a system meant to avoid false alarms, but also means some real borderline events — especially smaller night-only ones — go unflagged.

Dataset / stage	TP	FP	FN	Precision	Recall
A, stage 1	21	14	9	0.60	0.70
A, stage 2	16	0	14	1.00	0.53
B, stage 1	19	20	11	0.49	0.63
B, stage 2	12	0	18	1.00	0.40

TP, FP and FN are true positives, false positives and false negatives. A is the calibration dataset; B is the independently generated hold-out dataset.

Table 4.3: Warning performance per site-channel-day.

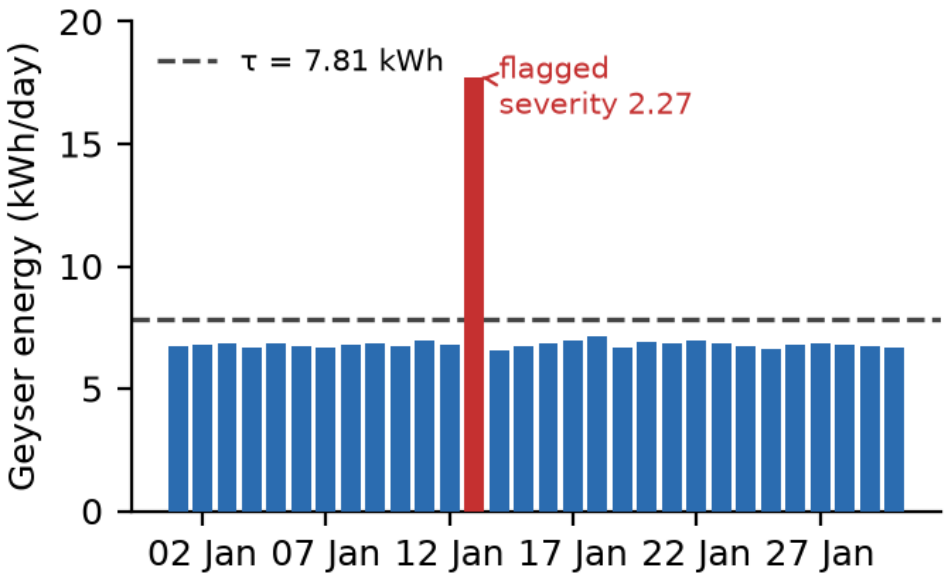


Figure 4.4: Daily geyser energy for Site HOUSE-004 with the stage 1 IQR threshold, computed live from the dashboard's own stored daily summaries; the injected event on 13 January is flagged (17.73 kWh against a threshold of 7.81 kWh, severity 2.27).

4.6 Performance and Responsiveness

Ingesting and validating 172,355 raw rows and aligning them to the grid took 1.3 s, and storage with daily summaries took a further 2.1 s, giving 3.9 s end to end and a 23.5 MB database. Median query times were 0.04 ms for the KPI cards, 0.37 ms for the daily trend, 0.32 ms for load ranking, 0.02 ms for community comparison and 0.06 ms for warnings. A one-day interval view took 200.6 ms because it reads raw readings. The dashboard showed zero horizontal overflow at all four tested viewport widths (1366, 768, 375 and 320 px), loaded locally in 41 ms, switched sites in about 12 ms and produced the CSV export correctly. These times exclude network latency.

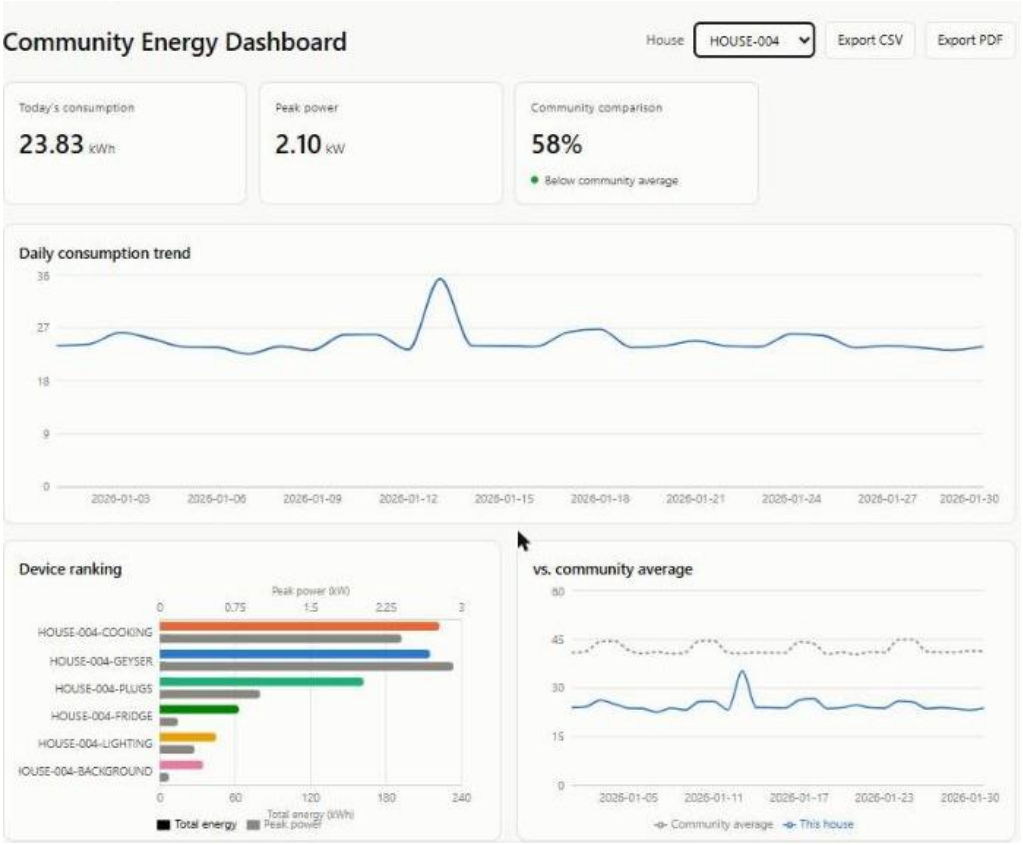


Figure 4.5: Dashboard overview for Site HOUSE-004: KPI cards, daily trend, device ranking and community comparison, from the live dashboard.

4.7 Validation Against Public Smart-Meter Data

4.7.1 External validity and comparison boundaries

An independent real-data source provides a useful challenge to a simulation-first architecture. It introduces household diversity and a different measurement origin without requiring the simulator's assumptions to hold. The reported successful run therefore strengthens the claim that the processing architecture is source-agnostic at the level tested. Nevertheless, the external dataset consists of a limited selection of London households, not a representative South African sample, and whole-house smart-meter readings do not provide the same circuit-level breakdown as a multi-channel controller.

External validation should consequently be interpreted at two levels. At the software level, the pipeline's ability to ingest and analyse a new source supports transferability of the data model and core calculations. At the behavioural level, the greater variance in the real sample exposes the limits of whole-community averages. These findings are compatible: the pipeline can generalise technically while a particular benchmark becomes less meaningful statistically. This distinction is a central contribution of the evaluation.

The pipeline and dashboard were also run, unmodified except for a new whole_house controller-channel type, against ten households from the Low Carbon London smart-meter trial (UK Power Networks, 2015), an open, CC-BY-licensed public dataset of half-hourly whole-house readings. A 30-day window common to all ten households (15 December 2011 to 14 January 2012) was ingested through the same validation, storage and analytics layers used for the simulated dataset. Because this source reports only whole-house active power at a 30-minute native interval, with no voltage, current or power factor fields, rules R4 and R6 were not applicable and were skipped, and the grid-timestamp check (R2) was configured to the source's own 30-minute interval rather than the simulator's 15-minute one.

Validation behaved correctly on genuinely external data: of 14,410 rows, 14,400 were valid and the remaining 10 were a duplicate reading at the same timestamp in every household, a systematic boundary artefact in the source file rather than injected noise, caught by rule R3 without any code change.

Real monthly totals ranged from 134.5 kWh to 839.0 kWh across the ten households, a 6.2× spread, compared with the simulated dataset's 2.5× spread across its three fixed usage-profile multipliers. With no usage-profile grouping available for these households, a whole-community comparison (Figures 4.6 and 4.7) produced ratios from 0.39 to 3.14: the highest-use household (Figure 4.6) reached 314% of the community average even though no peer group existed to judge it against. This real-data run supports two conclusions: the ingestion-to-analytics pipeline generalises to a data source it was not designed around, and within-group rather than whole-community comparison matters more, not less, as real household variance grows.

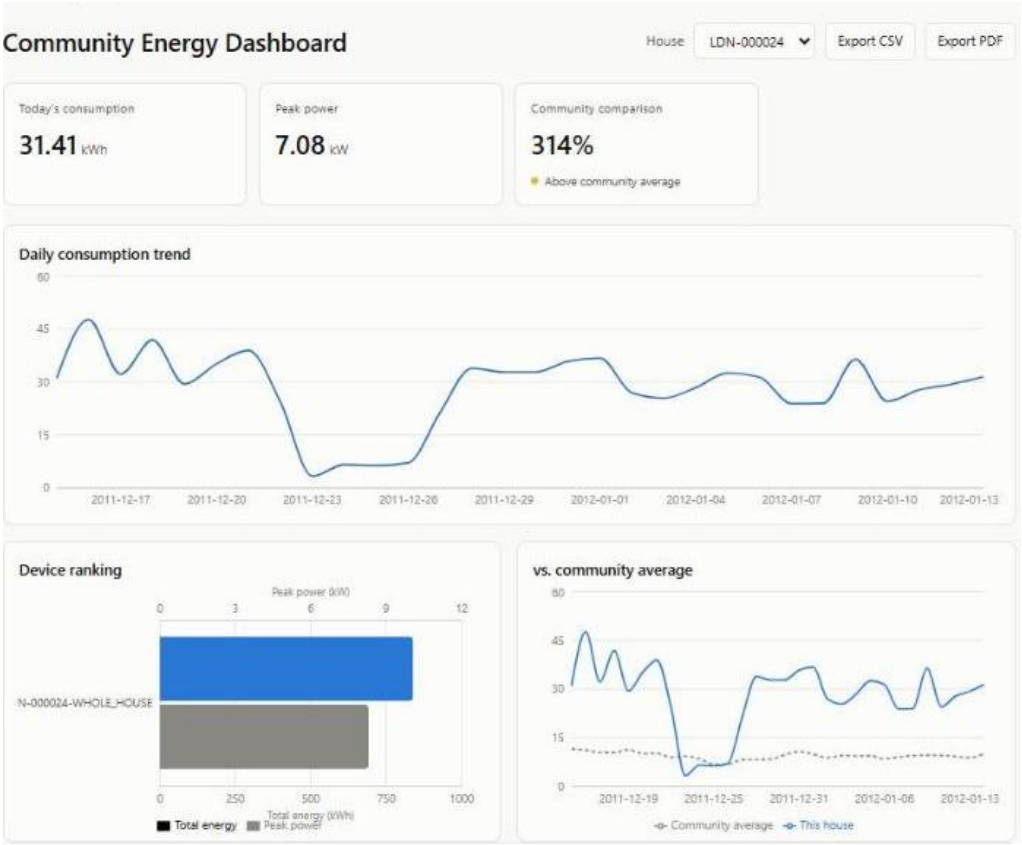


Figure 4.6: Dashboard for the highest-use real household (Low Carbon London, site LDN-000024), showing 314% of the community average with no peer-group comparison available.

HOUSEHOLD	TODAY'S KWH	PEAK KW	VS. COMMUNITY AVERAGE
LDN-000024	31.41	7.08	314% - Above average
LDN-000021	21.54	2.03	194% - Above average
LDN-000018	16.86	1.98	145% - Above average
LDN-000020	9.78	1.70	79% - Below average
LDN-000026	9.18	1.66	74% - Below average
LDN-000019	8.12	2.02	65% - Below average
LDN-000027	7.66	0.70	61% - Below average
LDN-000016	6.57	0.99	52% - Below average
LDN-000028	5.31	0.42	41% - Below average
LDN-000025	5.06	2.64	39% - Below average

Figure 4.7: Admin ranking view across the ten real Low Carbon London households, showing the ratio spread produced by whole-community comparison.

4.8 Compatibility with a Real Device Reading

A single reading from a Tuya-ecosystem smart plug (238 V, 551 mA, 87.6 W and 13.21 kWh cumulative) maps directly onto voltage_v, current_a, power_kw and cumulative_energy_kwh. Its implied power factor is 87.6 / (238 × 0.551) = 0.67. With the plug-channel assumption of 0.95, rule R6 gives a 29.5% deviation, well outside the 5% limit, so this reading would be flagged out_of_range. This is a defensible outcome for a prototype whose per-channel power factor is a single fixed assumption rather than a per-device measurement, since an off-the-shelf smart plug's real power factor depends on whatever appliance is plugged into it. Power factor should therefore be configurable per device when real device data is connected.

Chapter 5: Discussion, Limitations and Conclusion

5.1 Discussion

5.1.1 Synthesis across the research objectives

The completed work addresses the data-specification objective through a canonical representation of sites, channels, timestamps, electrical measurements and quality indicators. The pipeline objective is supported by the implemented validation, gap handling, relational storage and audit structures. The analytics objective is supported by independently tested energy, cost, peak, ranking, baseline, community, completeness and warning functions. The interface objective is supported by the responsive dashboard and export functions described in the implementation. The evaluation objective is partially complete: quantitative technical testing was conducted, but participant-based interpretation testing remains outstanding.

This synthesis is important because the project's success should not be judged by a single number. The 240 correctly rejected invalid records support validation-rule operation; the near-zero independent recomputation differences support arithmetic accuracy; the hold-out warning results disclose a genuine detection trade-off; and the real-data run demonstrates both architectural flexibility and the limitations of a simplistic peer benchmark. Together, these results support the narrower conclusion that a technically verified dashboard pipeline has been developed. They do not yet support the stronger behavioural claim that the dashboard improves users' interpretation relative to raw tables.

The validation layer and the calculations behaved as specified, and the results in Chapter 4 show where simple rules break down. A plain interquartile-range fence is sensitive but noisy, and it misses events that change the timing of use more than the daily total. Adding severity grading and a time-of-day rule fixed both problems in the hold-out test, at a clear and quantified cost in recall. Because both the calibration and hold-out datasets come from the same generator, the hold-out result demonstrates generalisation to new events, not to real households; the separate Low Carbon London validation (Section 4.7) is what demonstrates generalisation to a genuinely independent data source, and it did so successfully.

The real-data run also surfaced an honesty point that is central to this thesis rather than a minor caveat. The design in Chapter 3 specified a within-group peer comparison precisely to avoid the boomerang effect identified in Chapter 2's literature review (Schultz et al., 2007). That within-group comparison was not implemented in the live dashboard within the project period; the live system instead performs a whole-community, leave-one-out comparison. For the ten simulated households, whose consumption was deliberately generated within a narrow 2.5× spread, this difference is modest (Section 4.4: ratios of 0.57 to 1.61). For the ten real Low Carbon London households, whose consumption spans a genuine 6.2× spread, the same whole-community comparison produces a far more extreme and more misleading result (Section 4.7: ratios of 0.39 to 3.14, with the highest-use

household shown at 314% of the community average). This is reported directly rather than minimised: it is evidence that within-group comparison matters more, not less, as real household variance grows, and it is the strongest single argument in this thesis for prioritising that feature in future work over any other extension.

The administrator-only ranking view that exists in the delivered system is a separate mechanism from the resident-facing comparison, and the two should not be conflated. Residents see only a neutral status band tied to their own baseline, consistent with the boomerang-avoidance design decision in Chapter 3; the administrator ranking exists for operational oversight and is not shown to residents. Mixing simulated and real households in a single ranked view was found, during the course of this work, to produce ranking percentages that are not reproducible by a reader from the visible numbers alone, because the two populations are drawn from calendar periods that do not overlap. The administrator views shown in Chapter 4 are therefore reported separately for the simulated households (Figure 4.2) and the real households (Figure 4.7), rather than combined, which is both more honest and more useful as a diagnostic of the whole-community-comparison limitation discussed above.

5.2 Limitations

5.2.1 Threats to validity and reproducibility

Internal validity benefits from the known simulator ground truth, but the realism of generated load schedules remains a constraint. Construct validity is also relevant: a statistical abnormal-use event created by multiplying a channel's energy does not cover every operational fault a real controller may encounter. External validity is improved by the public-data run but remains limited by the sample's geography, measurement granularity and size. The absence of a participant study means that the interpretability hypothesis cannot be adjudicated from the technical tests alone.

Reproducibility would be strengthened by archiving the simulator configuration, random seeds, schema migrations, algorithm versions, validation rules and evaluation scripts together with a description of the runtime environment. Where public datasets have licensing or access conditions, a reproducibility package should provide lawful download instructions rather than redistributing restricted records. Future reporting should distinguish immutable experimental results from later software changes so that improvements to the dashboard do not retrospectively alter the evidence presented in this thesis.

Simulated data cannot show behaviour change or live communication reliability; only the real-data run (Section 4.7) speaks to the latter, and only indirectly.

Real faults such as clock drift, partial records and sensor offsets are messier than the faults injected under experimenter control in the simulated dataset.

The tariff used for cost estimation is flat and illustrative, whereas South African municipalities use block and time-of-use tariffs.

Thirty days is too short for a seasonal baseline.

The within-group (profile-matched) peer comparison specified in Chapter 3 was not implemented in the live dashboard; the delivered comparison is a whole-community average, which Chapter 4 shows becomes substantially more distorting as household variance grows.

The participant-based usability study and the System Usability Scale evaluation defined in Chapter 3 had not been conducted at the time of writing, so H1 and H0 (Chapter 1) cannot yet be formally accepted or rejected from user data; Chapter 4's results establish the technical precondition for that study rather than substituting for it.

The per-channel power factor used by the simulator and validation rules is a single fixed assumption per load category rather than a per-device measurement, which Section 4.8 shows causes a real device reading to be flagged even though the reading itself is plausible.

5.3 Recommendations

5.3.1 Prioritised engineering roadmap

The highest-priority analytical improvement is profile-matched peer comparison. A future implementation should define matching variables, minimum group sizes and the handling of sites without suitable peers before evaluating its effects on comparison stability. A second priority is a participant-based study using the already specified interpretation tasks and System Usability Scale. Participants should answer equivalent questions using raw readings and the dashboard, with task order and difficulty controlled to reduce learning effects. The resulting data would allow the research hypothesis to be tested directly rather than inferred from technical functionality.

Further engineering work should include an approved physical-controller integration, longer-duration datasets spanning seasons, robust tariff modelling and more systematic security testing. Warning algorithms could be compared at fixed false-alert budgets and across varying event magnitudes, while imputation policies could be tested separately for continuously operating and frequently switched loads. These are recommendations for future experiments; no additional results from them are claimed in the present thesis.

A concrete recommendation follows directly from the data-access limitation that shaped this project from its proposal stage. Rather than depending on vendor cloud access, each CBi Astute Smart Controller could be reflashed with custom, open firmware that exposes a lightweight local API and pushes its own readings straight to this system's ingestion endpoint, tagged by household. This would remove the per-site vendor-API approval that this project could not assume, let each household stream live data directly into the existing validation, storage and analytics layers unmodified, and keep the dashboard per-household rather than dependent on a shared cloud intermediary.

5.4 Conclusion and Future Work

A source-agnostic pipeline and dashboard were built and tested against a simulated smart-controller dataset with known ground truth. Validation caught every injected invalid record, the analytics matched independent recomputation to floating-point precision, short-gap interpolation reduced energy error to below 1.4% on average, and a two-stage warning rule reached full precision on unseen events at a recall cost that was measured rather than assumed. Pairing peer comparison with each household's own baseline, and reporting it as a neutral status band rather than a numeric leaderboard position, kept the resident-facing comparison from reading as a misleading community-wide ranking, even though the underlying average is drawn from the whole community rather than a matched peer group. The pipeline was additionally validated, unmodified, against ten real households from the public Low Carbon London smart-meter dataset, which confirmed real-data compatibility and sharpened the case for within-group comparison.

Future work, in priority order, is to:

- implement the within-group (profile-matched) peer comparison specified in Chapter 3, which Chapter 4's real-data results show is the single change most likely to improve the honesty of the resident-facing comparison;
- complete the System Usability Scale and interpretation study defined in Chapter 3, which is required to formally accept or reject the hypothesis stated in Chapter 1;
- reflash CBi Astute controllers with custom firmware exposing a local API, to stream live per-household readings directly to the dashboard without vendor-cloud dependency;
- extend real-data validation to South African load-research (DEL) data (Toussaint and Moodley, 2020), to test the pipeline against a dataset closer to the project's own national context than the UK-based Low Carbon London trial;
- add block and time-of-use tariffs to the cost-estimation algorithm, in place of the current flat illustrative tariff; and
- evaluate interpretable forecasting as an extension to the warning layer, drawing on the probabilistic load-forecasting (Hong and Fan, 2016) and model-explanation (Barredo Arrieta et al., 2020) literature, only once the core comparative dashboard and its peer-comparison feature are complete.

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